



NVIDIA Jetson Thor Module Carrier Board

Product Specification

Document History

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Version	Date	Description of Change
1.0	August 13, 2025	Initial release
1.1	September 10, 2025	Updated the following: > Added Microfit3.0 power connector reference to SCL > J87, J91, J502 not supported > Table 2-10 and Table 2-12
1.2	November 26, 2025	Updated the following: > Updated Figure 1-1 > Modified label for Microfit3.0 Power Connector > Figure 6-1, Table 6-1 and Table 6-2

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Chapter 1. Introduction

This specification contains recommendations and guidelines for engineers to follow to create modules for the expansion connectors on the NVIDIA® Jetson™ Thor Module Carrier Board, as well as understand the capabilities of the other dedicated interface connectors and associated power solutions on the platform.

**Caution:**

ALWAYS CONNECT THE JETSON THOR SERIES MODULE AND ALL EXTERNAL PERIPHERAL DEVICES BEFORE CONNECTING THE POWER SUPPLY TO THE DEV KIT.

The Jetson Thor Module Carrier Board contains ESD-sensitive parts. Always use appropriate anti-static and grounding techniques when working with the system. Failure to do so can result in ESD discharge to sensitive pins and irreparably damage your Jetson Thor Module Carrier Board. NVIDIA will not replace units that have been damaged due to ESD discharge.

The Jetson Thor Module Carrier Board connectors are used to access Jetson Thor features and interfaces. This enables a highly flexible and extensible development platform. Go to <https://developer.nvidia.com/embedded-computing> or contact your NVIDIA representative for access to software updates and the developer SDK that supports the OS image and host development platform that you want to use. The developer SDK includes an OS image that you will load onto your Jetson Thor device, supporting documentation, and code samples to help you get started.



Note: References to Dev Kit apply to Jetson Thor Module Carrier Board except where explicitly noted.

1.1 Jetson Thor Series Feature List

The following is a list of features for the Jetson Thor Series module.

- > Applications processor
 - NVIDIA Thor™
- > Memory
 - LPDDR5X DRAM
- > Advanced power management
 - Dynamic voltage and frequency scaling
 - Multiple clock and power domains
 - Thermal transfer plate (TTP) and optional fan or heat sink

1.2 Carrier Board Feature List

The following is a list of features for the carrier board.

- > Connection to Jetson Thor
 - 699-pin (11 × 65) board-board connector
- > USB
 - 2× USB 10Gbps Type-A connector
 - 2× USB 5Gbps Type-C connector
- > Wired Network
 - MGBE: Up to 5Gb/s to RJ45 connector
- > Display
 - VESA® DisplayPort® (DP) output connector
 - HDMI 2.1 output connector
- > QSFP28 connector
 - 4× MGBE (XFI)
- > M.2 Key E Connector
 - PCIe ×1 Lane, USB 2.0
 - I2S, UART Control
- > M.2 Key M Connector supporting NVMe for storage
 - PCIe ×4 Lane, I2C (J103)
- > CAN Header
 - 2x CAN port
- > User Interfaces and Indicators
 - Automation Header
 - Power, Reset, Force Recovery, Sleep, Overcurrent Throttling Enable, Auto-Power-ON Enable

- LED: Main 5.0V Supply
- > Debug
 - USB Type-C® (USB-C®) Debug Connector
- > Power
 - 2× USB-C (either for power input – Power Adapter included)
 - Microfit3.0 Connector: 9V to 28V (alternative to USB-C).
 - Main 5V Supply: TPS53015
 - Main 3.3V Supply: TPS53015
 - Main 1.8V Buck Supply: MP2384
 - USB Type-A VBUS Load Switches: AP22811
 - 12V Buck-Boost: MP28167GQ
 - Load Switches/LDOs (DP, HDMI)
- > Operating Temperature Range
 - 0 °C to 35 °C



Note: If there is more than one power supply (e.g., Microfit3.0 or USB Type-C) connected, then the USB PD Controller (CYPD8225) will let the first power supply through, also known as First come first serve.

The USB 3.2 specification absorbed all prior 3.x specifications. USB 3.2 identifies three transfer rates – 20Gbps (Gen 2x2), 10Gbps (Gen 1x2 or Gen 2x1) and 5Gbps (Gen 1x1).

For the Microfit3.0 part number information, refer to the latest version of the *Jetson Thor Series Supported Component List* (DA-12429-001), available for download on the Jetson Download center.

1.3 Jetson Thor Module Carrier Board Block Diagram

Figure 1-1 through Figure 1-3 show the block diagram and various placement views for Jetson Thor and the carrier board.

Figure 1-1. Jetson Thor Module Carrier Board Block Diagram

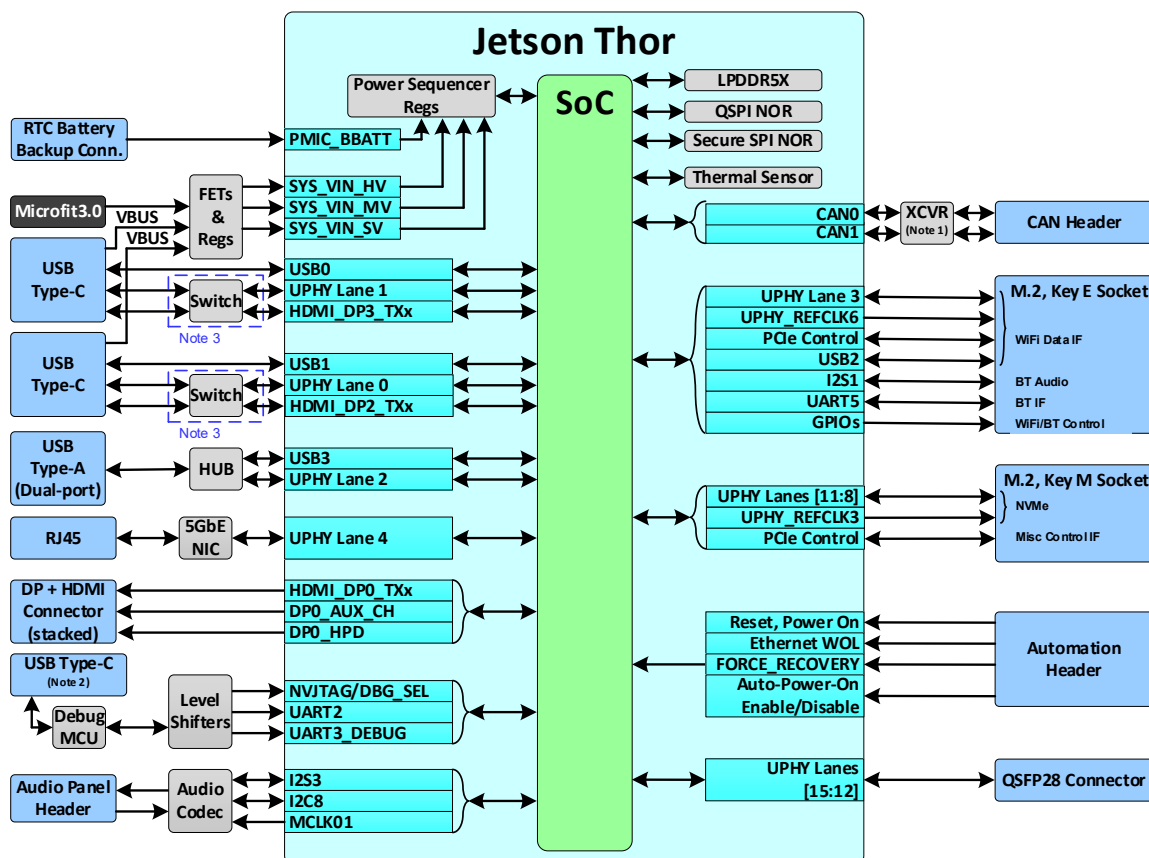


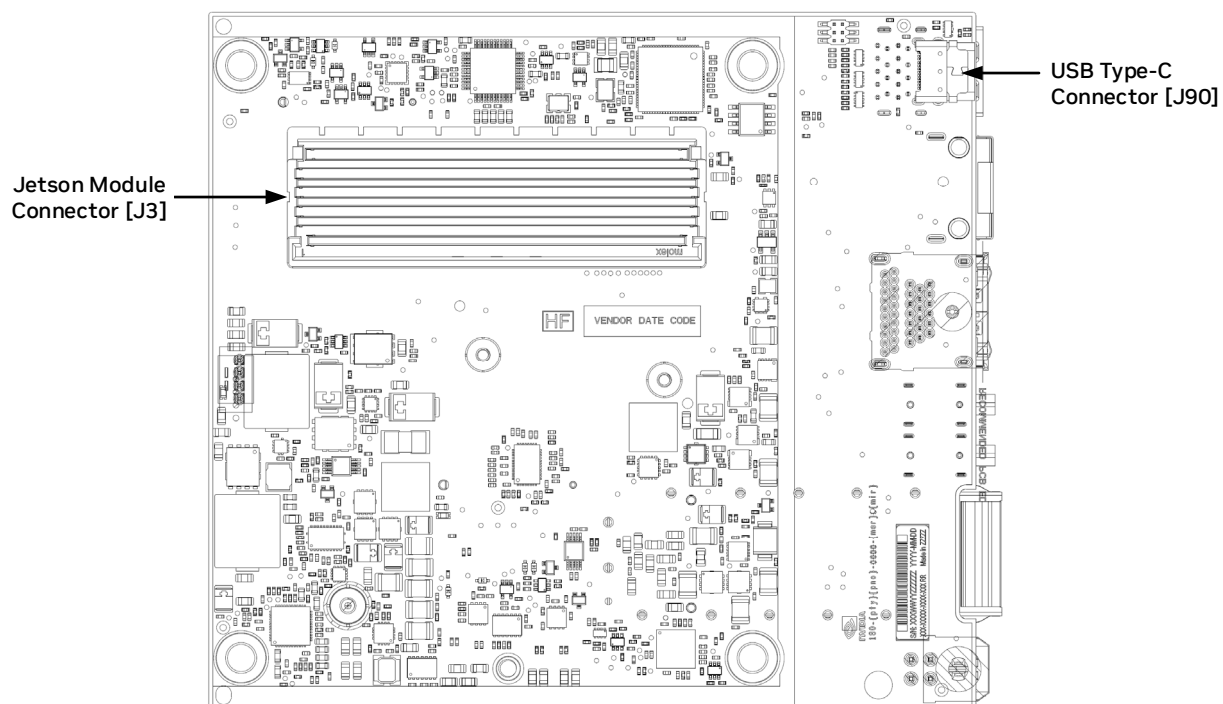
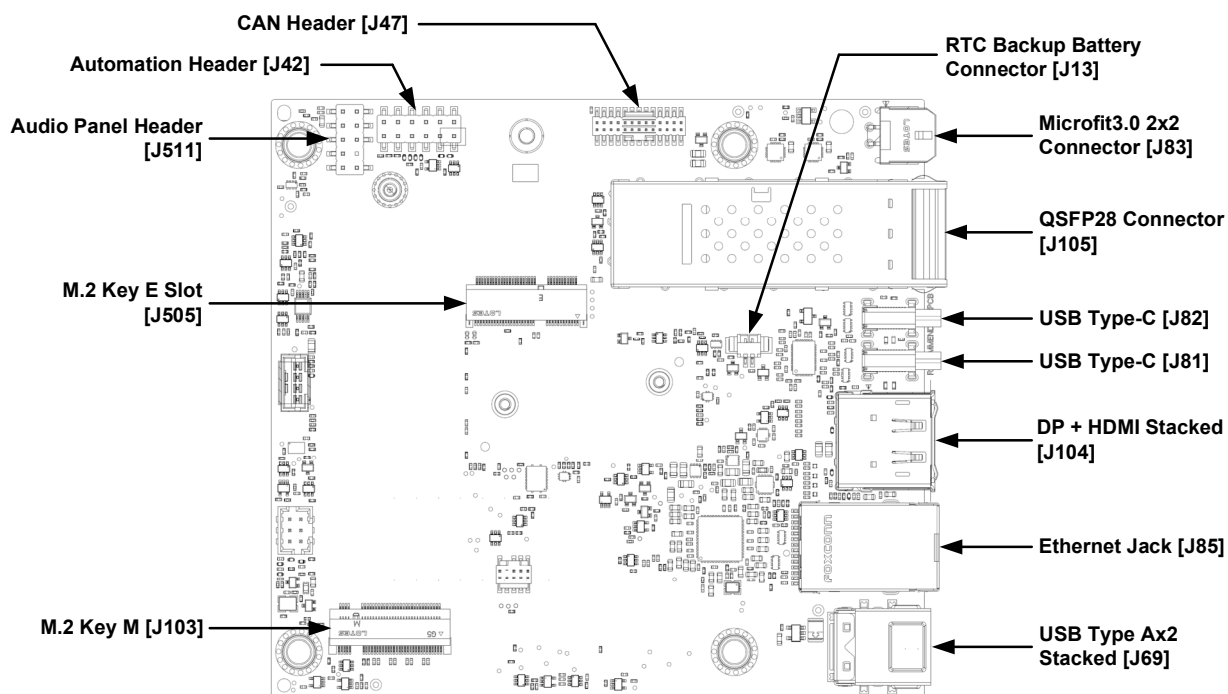
Figure 1-2. Carrier Board Placement – Module Connector Side**J3** Jetson Module Connector**J90** USB Type-C Connector (Debug)

Figure 1-3. Carrier Board Placement – Non-Module Connector Side



J13	RTC Battery Backup Connector	J85	Ethernet Jack
J42	Automation Header	J103	M.2 Key M Connectivity Slot (NVMe is preinstalled)
J47	CAN Header	J104	DP + HDMI Stacked
J69	USB Type-A Stacked	J105	QSFP28 Connector
J81	USB Type-C Connector #1	J505	M.2 Key E Connectivity Slot (WiFi/BT module is preinstalled)
J82	USB Type-C Connector #2	J511	Audio Panel Header
J83	Microfit3.0 2x2 Connector		

Chapter 2. Carrier Board Standard Connectors

The Jetson Thor Module Carrier Board provides several standard expansion connectors to support additional functionality beyond what is integrated on the main platform board. This includes the following:

- > USB 10Gbps Type-A: 2× (Host Mode Gen 2 and Device Mode Gen 1)
- > USB 5Gbps Type-C: 2× (data transfer, power supply)
- > USB Type-C (debug purposes only)
- > RJ45 Connector
- > DP + HDMI Stacked Connector
- > QSFP28 Connector
- > M.2, Key E Slot
- > M.2, Key M Slot
- > Audio Panel Header

2.1 USB Ports

The carrier board has the following USB connectors:

- > 2x USB Type-C Connectors (J81 and J82). Both support USB 3.2 Gen 1x1, USB 2.0, and DP. J81 supports host and device modes as well as recovery mode. J82 supports host and device modes. VBUS supports up to 5A per USB-C connector.
Note: Only one of the USB-C connectors can support Device mode at a time.
- > 2x USB Type-A connector (J69). The USB Type-A connector support host mode for USB 3.2 Gen 2x1 and USB 2.0. VBUS for the connector support up to 2A, which is shared between the two ports. The bandwidth is also shared between the two ports.
- > USB Type-C connector (J90). Used only for debug purposes.

2.1.1 USB Type-C

Table 2-1 and Table 2-2 list the USB 3.2 Gen 1x1 Type-C connector pin descriptions for J81 and J82.

Table 2-1. USB 3.2 Type-C Connector Pin Descriptions for J81

Pin #	Connector Pin Name	Associated Module Pin Name (see note 1)	Module Pin #	Usage/Description	Type/Dir
A1	GND_A	–	–	Ground	Ground
A2	TX1_P	UPHY_TX1_P	J22	USB 3.2 – Transmit 1 from Mux	Output
A3	TX1_N	UPHY_TX1_N	J23		
A4		–	–	USB VBUS_A Power	Power
A5	CC1	–	–	CC 1 from CC Controller	Bidir
A6	D1_P	USB0_P	F12	USB 2.0 #0 Data 1	Bidir
A7	D1_N	USB0_N	F13		
A8	SBU1	DP_AUX_CH3_P	F20	Unused	Bidir
A9		–	–	USB VBUS_A Power	Power
A10	RX2_N	UPHY_RX1_N	A23	USB 3.2 – Receive 2 from Mux	Input
A11	RX2_P	UPHY_RX1_P	A22		
A12	GND_A	–	–	Ground	Ground
B1	GND_B	–	–	Ground	Ground
B2	TX2_P	UPHY_TX1_P	J22	USB 3.2 – Transmit 2 from Mux	Output
B3	TX2_N	UPHY_TX1_N	J23		
B4		–	–	USB VBUS_A Power	Power
B5	CC2	–	–	CC 2 from CC Controller	Bidir
B6	D2_P	USB0_P	F12	USB 2.0 #0 Data 2	Bidir
B7	D2_N	USB0_N	F13		
B8	SBU2	DP_AUX_CH3_N	F21	Unused	Bidir
B9		–	–	USB VBUS_A Power	Power
B10	RX1_N	UPHY_RX1_N	A23	USB 3.2 – Receive 1 from Mux	Input
B11	RX1_P	UPHY_RX1_P	A22		
B12	GND_B	–	–	Ground	Ground

Notes:

1. The module pins for the USB 3.2 ports are not directly connected to the USB connector pins but are routed through a switch.
2. In the Type/Dir column, Output is to USB connectors; Input is from USB connectors. Bidir is for bidirectional signals.

Legend

Ground	Power	Reserved/Unused
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Table 2-2. USB 3.2 Type-C Connector Pin Descriptions for J82

Pin #	Connector Pin Name	Associated Module Pin Name (see note 1)	Module Pin #	Usage/Description	Type/Dir
A1	GND_A	–	–	Ground	Ground
A2	TX1_P	UPHY_TX0_P	K32	USB 3.2 – Transmit 1 from Mux	Output
A3	TX1_N	UPHY_TX0_N	K33		
A4	–	–	–	USB VBUS_B Power	Power
A5	CC1	–	–	CC 1 from CC Controller	Bidir
A6	D1_P	USB1_P	C11	USB 2.0 #0 Data 1	Bidir
A7	D1_N	USB1_N	C10		
A8	SBU1	DP_AUX_CH2_P	E19	Unused	Bidir
A9	–	–	–	USB VBUS_B Power	Power
A10	RX2_N	UPHY_RX0_N	C35	USB 3.2 – Receive 2 from Mux	Input
A11	RX2_P	UPHY_RX0_P	C34		
A12	GND_A	–	–	Ground	Ground
B1	GND_B	–	–	Ground	Ground
B2	TX2_P	UPHY_TX0_P	K32	USB 3.2 – Transmit 2 from Mux	Output
B3	TX2_N	UPHY_TX0_N	K33		
B4	–	–	–	USB VBUS_B Power	Power
B5	CC2	–	–	CC 2 from CC Controller	Bidir
B6	D2_P	USB1_P	C11	USB 2.0 #0 Data 2	Bidir
B7	D2_N	USB1_N	C10		
B8	SBU2	DP_AUX_CH2_N	E18	Unused	Bidir
B9	–	–	–	USB VBUS_B Power	Power
B10	RX1_N	UPHY_RX0_N	C35	USB 3.2 – Receive 1 from Mux	Input
B11	RX1_P	UPHY_RX0_P	C34		
B12	GND_B	–	–	Ground	Ground

Notes:

1. The module pins for the USB 3.2 ports are not directly connected to the USB connector pins but are routed through a switch.
2. In the Type/Dir column, Output is to USB connectors; Input is from USB connectors. Bidir is for bidirectional signals.

Legend

Ground	Power	Reserved/Unused
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2.1.2 USB Type-A Dual-Port Connector from Hub

Table 2-3 lists the USB 3.2 Gen 2x1 Dual-port Type-A connector pin descriptions for J69.

Table 2-3. USB 3.2 Type A Dual-port Connector Pin Descriptions for J69

Pin #	Connector Pin Name (Note 1)	Associated Module Pin Name (Note 2)	Module Pin #	Usage/Description	Type/Dir
USB 3.2 Type-A (0)					
1	VBUS	–	–	VBUS Supply	Power
2	DN_0 (DSP2)	USB3_N	G10	USB 2.0 #3 Data (stack connector 2)	Bidir
3	DP_0 (DSP2)	USB3_P	G11		
4	GND	–	–	Ground	Ground
5	RX_N_0 (DSP2)	UPHY_RX2_N	C22	USB 3.2 (SoC Port 2) Receive (stack connector 2)	Input
6	RX_P_0 (DSP2)	UPHY_RX2_P	C23		
7	GND	–	–	Ground	Ground
8	TX_N_0 (DSP2)	UPHY_TX2_N	G22	USB 3.2 (SoC Port 2) Transmit (stack connector 2)	Output
9	TX_P_0 (DSP2)	UPHY_TX2_P	G23		
USB 3.2 Type-A (1)					
10	VBUS	–	–	VBUS Supply	Power
11	DN_1 (DSP1)	USB3_N	G10	USB 2.0 #3 Data (stack connector 1)	Bidir
12	DP_1 (DSP1)	USB3_P	G11		
13	GND	–	–	Ground	Ground
14	RX_N_1 (DSP1)	UPHY_RX2_N	C22	USB 3.2 (SoC Port 2) Receive (stack connector 1)	Input
15	RX_P_1 (DSP1)	UPHY_RX2_P	C23		
16	GND	–	–	Ground	Ground
17	TX_N_1 (DSP1)	UPHY_TX2_N	G22	USB 3.2 (SoC Port 2) Transmit (stack connector 1)	Output
18	TX_P_1 (DSP1)	UPHY_TX2_P	G23		

Notes:

1. These signals are from the Hub.
2. These are from the Jetson Thor to the Hub.
3. In the Type/Dir column, Output is to USB connectors; Input is from USB connectors. Bidir is for bidirectional signals.

Legend

Ground	Power
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2.1.3 USB Type-C (Debug)

A USB Type-C connector is provided for debug purposes only. The connector USB 2.0 interface comes from a debug MCU. It takes in the following interfaces and control pins that can be used for debug and development:

- > UART3_TX/RX_DEBUG
- > UART2_TX/RX

Table 2-4. USB Type-C Debug Connector Pin Descriptions for J90

Pin #	Connector Pin Name	Associated Module Pin Name	Module Pin #	Usage/Description	Type/Dir
A1	GND	–	–	Ground	Ground
A2	TX1_P	–	–	No Connect	–
A3	TX1_N	–	–	No Connect	–
A4	VBUS_A	–	–	VBUS Supply	Power
A5	CC1	–	–	4.7kΩ PD to GND	Ground
A6	D1_P	–	–	USB 2.0 Data (From Debug MCU)	Bidir
A7	D1_N	–	–		
A8	SBU1	–	–	No Connect	–
A9	VBUS_A	–	–	VBUS Supply	Power
A10	RX2_N	–	–	No Connect	–
A11	RX2_P	–	–	No Connect	–
A12	GND	–	–	Ground	Ground
B1	GND	–	–	Ground	Ground
B2	TX2_P	–	–	No Connect	–
B3	TX2_N	–	–	No Connect	–
B4	VBUS_B	–	–	VBUS Supply	Power
B5	CC2	–	–	4.7kΩ PD to GND	Ground
B6	D2_P	–	–	USB 2.0 Data (From Debug MCU)	Bidir
B7	D2_N	–	–		
B8	SBU2	–	–	No Connect	–
B9	VBUS_B	–	–	VBUS Supply	Power
B10	RX1_N	–	–	No Connect	–
B11	RX1_P	–	–	No Connect	–
B12	GND	–	–	Ground	Ground

Note: In the Type/Dir column, Output is to USB connectors; Input is from USB connectors. Bidir is for bidirectional signals.

Legend

Ground	Power	Reserved/Unused
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2.2 Ethernet Connector

The carrier board implements an RJ45 connector (J85) with built-in magnetics to support 5 Gigabit Ethernet (GbE). Data rates supported include 1 Gb, 2.5 Gb, and 5 Gb.

Table 2-5. PCIe NIC RJ45 Connector Pin Description

Pin #	Connector Pin Name	Associated Module Pin Name (Note 1)	Module Pin #	Usage/Description	Type/Dir
R1	CT1	–	–	Center tap 1	–
R2	TRD1+	UPHY_TX4_P UPHY_RX4_P	K21 B21	MDI pair A	Bidir
R3	TRD1-	UPHY_TX4_N UPHY_RX4_N	K20 B20		
R4	TRD2+	UPHY_TX4_P UPHY_RX4_P	K21 B21	MDI pair B	Bidir
R5	TRD2-	UPHY_TX4_N UPHY_RX4_N	K20 B20		
R6	TRD3+	UPHY_TX4_P UPHY_RX4_P	K21 B21	MDI pair C	Bidir
R7	TRD3-	UPHY_TX4_N UPHY_RX4_N	K20 B20		
R8	TRD4+	UPHY_TX4_P UPHY_RX4_P	K21 B21	MDI pair D	Bidir
R9	TRD4-	UPHY_TX4_N UPHY_RX4_N	K20 B20		
R10	CT_GND	–	–	Ground	Ground
L1	GRN- (Right)	–	–	Right green LED cathode	Input
L2	OG- (Right)	–	–	Right yellow LED cathode	Input
L3	GRN- (Left)	–	–	Left green LED cathode	Input
L4	GRN+ (Left)	–	–	Left green LED common (anode)	Power

Notes:

1. The module pins are not directly connected to the RJ45 connector pins but are routed through a PCIe NIC.
2. In the Type/Dir column, Output is to RJ45 connector; Input is from RJ45 connector. Bidir is for bidirectional signals.

Legend

Ground	Power	Reserved/Unused
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2.3 DisplayPort + HDMI Stacked Connector

2.3.1 DisplayPort Connector

A standard DisplayPort (DP) connector is supported on the stacked connector (J104).

Table 2-6. DisplayPort Connector Pin Description

Pin #	Connector Pin Name	Associated Module Pin Name	Module Pin #	Usage/Description	Type/Dir
1	LANE_0P	DP0_D0_HDMI_D2_P	D51	DP Lane 0+	Output
2	GND	–	–	Ground	Ground
3	LANE_0N	DP0_D0_HDMI_D2_N	D52	DP Lane 0–	Output
4	LANE_1P	DP0_D1_HDMI_D1_P	B51	DP Lane 1+	Output
5	GND	–	–	Ground	Ground
6	LANE_1N	DP0_D1_HDMI_D1_N	B52	DP Lane 1–	Output
7	LANE_2P	DP0_D2_HDMI_D0_P	A51	DP Lane 2+	Output
8	GND	–	–	Ground	Ground
9	LANE_2N	DP0_D2_HDMI_D0_N	A50	DP Lane 2–	Output
10	LANE_3P	DP0_D3_HDMI_CK_P	C51	DP Lane 3+	Output
11	GND	–	–	Ground	Ground
12	LANE_3N	DP0_D3_HDMI_CK_N	C50	DP Lane 3–	Output
13	MODE	–	–	MODE: Selects between DP and TMDS (DVI/HDMI) signaling.	Unused
14	CEC	–	–	Unused	–
15	AUXP	DP_AUX_CH0_P	G53	DisplayPort Auxiliary Channel 0+	Bidir
16	GND	–	–	Ground	Ground
17	AUXN	DP_AUX_CH0_N	G54	DisplayPort Auxiliary Channel 0–	Bidir
18	HPD	HPD0	K50	DP Hot Plug Detect	Input
19	PWR_RET	–	–	Power Return (Ground)	Ground
20	PWR	–	–	+3.3V	Power

Pin #	Connector Pin Name	Associated Module Pin Name	Module Pin #	Usage/Description	Type/Dir
Note: In the Type/Dir column, Output is to DP connector; Input is from DP connector. Bidir is for bidirectional signals.					

Legend	Ground	Power	Reserved/Unused
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2.3.2 HDMI Connector

A standard HDMI connector is supported on the stacked connector (J104).

Table 2-7. HDMI Connector Pin Description

Pin #	Connector Pin Name	Associated Module Pin Name	Module Pin #	Usage/Description	Type/Dir
1	D2+	DP1_D0_HDMI_D2_P	H49	HDMI Lane 2+	Output
2	GND	-	-	Ground	Ground
3	D2-	DP1_D0_HDMI_D2_N	H48	HDMI Lane 2-	Output
4	D1+	DP1_D1_HDMI_D1_P	G51	HDMI Lane 1+	Output
5	GND	-	-	Ground	Ground
6	D1-	DP1_D1_HDMI_D1_N	G50	HDMI Lane 1-	Output
7	D0+	DP1_D2_HDMI_D0_P	J47	HDMI Lane 0+	Output
8	GND	-	-	Ground	Ground
9	D0-	DP1_D2_HDMI_D0_N	J48	HDMI Lane 0-	Output
10	CK+	DP1_D3_HDMI_CK_P	K46	HDMI CLK+	Output
11	GND	-	-	Ground	Ground
12	CK-	DP1_D3_HDMI_CK_N	K47	HDMI CLK-	Output
13	CEC_HDMI	HDMI_CEC	-	HDMI CEC	Bidir
14	Reserved	-	-	Unused	-
15	SCL	DP_AUX_CH1_P	D16	HDMI DDC+	Bidir
16	SDA	DP_AUX_CH1_N	D17	HDMI DDC-	Bidir
17	GND	-	-	Ground	Ground
18	+5V	-	-	+5V	Power
19	HP_DET	HPD1	C18	HDMI Hot Plug Detect	Input

Note: In the Type/Dir column, Output is to HDMI connector. Input is from HDMI connector. Bidir is for bidirectional signals.

Legend	Ground	Power	Reserved/Unused
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2.4 QSFP28 Connector

The carrier board includes a QSFP28 connector (J105) that makes accessible several critical system control signals. The connector used on the carrier board is a TE Connectivity Ltd. 1551920-2.

Table 2-8. QSFP28 Connector Pin Description

Pin #	Connector Pin Name	Associated Module Pin Name	Module Pin #	Usage/Description	Type/Dir
1	GND	–	–	Ground	Ground
2	TX2N	UPHY_TX13_N	K28	MGBE C1 (XFI)	Output
3	TX2P	UPHY_TX13_P	K29	MGBE C1 (XFI)	Output
4	GND	–	–	Ground	Ground
5	TX4N	UPHY_TX15_N	J31	MGBE C3 (XFI)	Output
6	TX4P	UPHY_TX15_P	J30	MGBE C3 (XFI)	Output
7	GND	–	–	Ground	Ground
8	MODSEL*	–	–	Ground	Ground
9	RESET*	SOC_QSFP_RESET	J52	Reset	Output
10	VCC_RX	–	–	+3.3V	Power
11	SCL	I2C3_CLK	F53	I2C CLK	Bidir
12	SDA	I2C3_DAT	E53	I2C DATA	Bidir
13	GND	–	–	Ground	Ground
14	RX3P	UPHY_RX14_P	C31	MGBE C2 (XFI)	Input
15	RX3N	UPHY_RX14_N	C30	MGBE C2 (XFI)	Input
16	GND	–		Ground	Ground
17	RX1P	UPHY_RX12_P	D28	MGBE C0 (XFI)	Input
18	RX1N	UPHY_RX12_N	D29	MGBE C0 (XFI)	Input
19, 20	GND	–		Ground	Ground
21	RX2N	UPHY_RX13_N	B28	MGBE C1 (XFI)	Input
22	RX2P	UPHY_RX13_P	B29	MGBE C1 (XFI)	Input
23	GND	–		Ground	Ground
24	RX4N	UPHY_RX15_N	A31	MGBE C3 (XFI)	Input
25	RX4P	UPHY_RX15_P	A30	MGBE C3 (XFI)	Input
26	GND	–		Ground	Ground
27	MODPRS*	GPIO62	H17		Input
28	INT*	I2C0_DAT	J53		Input
29	VCC_TX	–	–	+3.3V	Power
29	VCC1	–	–	+3.3V	Power

Pin #	Connector Pin Name	Associated Module Pin Name	Module Pin #	Usage/Description	Type/Dir
31	LPMODE	GPIO57	G18		Output
32	GND	–		Ground	Ground
33	TX3P	UPHY_TX14_P	G31	MGBE C2 (XFI)	Output
34	TX3N	UPHY_TX14_N	G30	MGBE C2 (XFI)	Output
35	GND	–		Ground	Ground
36	TX1P	UPHY_TX12_P	H28	MGBE C0 (XFI)	Output
37	TX1N	UPHY_TX12_N	H29	MGBE C0 (XFI)	Output
38	GND	–		Ground	Ground

Note: In the Type/Dir column, Output is to QSFP28 connector. Input is from QSFP28 connector. Bidir is for bidirectional signals.

Legend

Ground	Power
--------	-------

2.5 M.2 Key E Expansion Slot

The carrier board includes an M.2, Key E slot (J505). This includes interface options for WLAN/BT including PCIe (x1), USB 2.0, UART, I2C, and I2S.

Table 2-9. M.2 Key E Expansion Slot Pin Description

Pin #	Connector Pin Name	Assoc. Module Pin Name	Module Pin #	Usage/Description	Type/Dir	Voltage Level
Odd Pins						
1	GND	–	–	Ground	GND	–
3	USB_DP	USB2_P	A10	USB 2.0 #2	Bidir	USB
5	USB_DM	USB2_N	A11			
7	GND	–	–	Ground	GND	–
9, 11, 13, 15, 17, 19, 21, 23	NC	–	–	Unused	–	–
25, 27, 29, 31	Key	–	–	–	–	–
33	GND	–	–	Ground	GND	–
35	AP_PETP0	UPHY_TX3_P	G35	PCIe Transmit (C1)	Output	PCIe
37	AP_PETN0	UPHY_TX3_N	G34			
39	GND	–	–	Ground	GND	–
41	AP_PERP0	UPHY_RX3_P	B33	PCIe Receive (C1)	Input	PCIe
43	AP_PERN0	UPHY_RX3_N	B32			

Pin #	Connector Pin Name	Assoc. Module Pin Name	Module Pin #	Usage/Description	Type/Dir	Voltage Level
Odd Pins						
45	GND	–	–	Ground	GND	–
47	REFCLKP0	UPHY_REFCLK6_P	D49	PCIe Reference Clock (C1)	Output	PCIe
49	REFCLKN0	UPHY_REFCLK6_N	D48			
51	GND	–	–	Ground	GND	–
53	CLKREQ0#	PEX_C1_CLKREQ_N	B37	PCIe Clock Request (C1)	Bidir	3.3V
55	PEWAKE0#	GPIO21	B8	PCIe Wake (see note 1)	Output	3.3V
57	GND	–	–	Ground	GND	–
59	AP_RESERVED/PET P1	GPIO60	K6	SAR TOUT	Output	
61	AP_RESERVED/PET N1	GPIO33	C54	AP Wake BT	Output	
63	GND	–	–	Ground	GND	–
65, 67	NC	–	–	Unused	–	–
69	GND	–	–	Ground	GND	–
71, 73	NC	–	–	Unused	–	–
75	GND	–	–	Ground	GND	–
76, 77	MNT1, MNT2	–	–	Ground	GND	–
Even Pins						
2, 4	3P3V	–	–	3.3V Supply	PWR	–
6	LED1#	–	–	Unused	–	–
8	PCM_CLK/12S SCK	I2S1_CLK	L14	I2S Clock	Bidir	
10	PCM_SYNC/12S WS	I2S1_FS	D8	I2S Frame/Word Select	Bidir	
12	AP_PCM_IN/12S SD_IN	I2S1_SDIN	H8	I2S Data In	Input	
14	AP_PCM_OUT/12S SD_OUT	I2S1_SDOUT	C7	I2S Data Out	Output	
16	LED2#	–	–	Unused	–	–
18	GND	–	–	Ground	GND	–
20	UART_WAKE#	I2C9_DAT	J55	BT Wake SoC Primary	Output	
22	AP_UART_RXD	UART5_RX	H58	UART Receive	Input	

Pin #	Connector Pin Name	Assoc. Module Pin Name	Module Pin #	Usage/Description	Type/Dir	Voltage Level
Odd Pins						
24, 26, 28, 30	Key	–	–	–	–	–
32	AP_UART_TXD	UART5_TX	J58	UART Transmit	Output	
34	AP_UART_CTS	UART5_CTS	H57	UART Clear to Send	Input	
36	AP_UART_RTS	UART5_RTS	K58	UART Ready to Send	Output	
38, 40	NC	–	–	Unused	–	–
42	VENDOR_DEFINED	GPIO33	C54	AP BT Wake	Output	1.8V
44, 46, 48	NC	–	–	Unused	–	–
50	SUSCLK_32KHZ	–	–	Unused	–	–
52	PERST0#	PEX_C1_RST_N	B36	PCIe Reset (C1)	Output	
54	W_DISABLE2#	UART4_CTS	L6	M2E BT Disable	Output	
56	W_DISABLE1#	GPIO01	J4	M2E Wi-Fi Disable	Output	
58	I2C_DATA	I2C8_DAT	F51	I2C 8 Clock (see note 2)	Bidir	
60	I2C_CLK	I2C8_CLK	F52	I2C 8 Data (see note 2)	Bidir	
62	ALERT#	GPIO48	D5	Alert to SoC	Input	1.8V
64, 66, 68, 70	NC	–	–	Unused	–	–
72, 74	3P3V	–	–	3.3V Supply	PWR	–

Notes:

1. Series resistor R1019 are unstuffed by default disconnecting the PCIe Wake signal from the M.2 Key E connector.
2. Series resistors R842 & R843 are unstuffed by default disconnecting the I2C signals from the M.2 Key E connector.
3. In the Type/Dir column, Output is to M.2 connector; Input is from M.2 connector. Bidir is for bidirectional signals.

Legend	Ground	Power	Reserved/Unused
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2.5.1 Maximum M.2 Key E Card Routing Delays

The routing on the carrier board meets the requirements in the *NVIDIA Jetson Thor Series Modules Design Guide* for routing to an M.2 connector. Therefore, the maximum allowed routing on the M.2 adapter card is based on the insertion loss allocated for the card which is 9.6 dB for Gen3 or 8.0 dB for Gen4. Using the mid-loss PCB technology loss numbers from the design guide, this translates to the values listed in the following table.

Table 2-10. M.2 Key E Card Maximum Trace Delays

PCIe	Maximum Trace Delay Allowed on M.2 Card
Gen3	127 mm (average of Microstrip and Stripline guidelines)
Gen4	

2.6 M.2 Key M Expansion Slot

The carrier board includes an M.2 Key M expansion slot (J103). It can be configured for PCIe (×4) for NVMe. The connector is Lotes Co, Ltd APCIA178-P003B06C part.

Table 2-11. M.2 Key M Expansion Slot Pin Description (J103)

Pin #	Connector Pin Name	Assoc. Module Pin Name	Module Pin #	Usage/Description	Type/Dir	Voltage Level
Odd Pins						
1	GND	–	–	Ground	GND	–
3	GND	–	–	Ground	GND	–
5	PER3_N	UPHY_RX11_N	A27	PCIe Receive Lane 3 (C5)	Input	PCIe
7	PER3_P	UPHY_RX11_P	A26			
9	GND	–	–	Ground	GND	–
11	PET3_N	UPHY_TX11_N	J27	PCIe Transmit Lane 3 (C5)	Output	PCIe
13	PET3_P	UPHY_TX11_P	J26			
15	GND	–	–	Ground	GND	–
17	PER2_N	UPHY_RX10_N	C26	PCIe Receive Lane 2 (C5)	Input	PCIe
19	PER2_P	UPHY_RX10_P	C27			
21	GND	–	–	Ground	GND	–
23	PET2_N	UPHY_TX10_N	G26	PCIe Transmit Lane 2 (C5)	Output	PCIe
25	PET2_P	UPHY_TX10_P	G27			
27	GND	–	–	Ground	GND	–
29	PER1_N	UPHY_RX9_N	B24	PCIe Receive Lane 1 (C5)	Input	PCIe

Pin #	Connector Pin Name	Assoc. Module Pin Name	Module Pin #	Usage/Description	Type/Dir	Voltage Level
Odd Pins						
31	PER1_P	UPHY_RX9_P	B25			
33	GND	–	–	Ground	GND	–
35	PET1_N	UPHY_TX9_N	K24	PCIe Transmit Lane 1 (C5)	Output	PCIe
37	PET1_P	UPHY_TX9_P	K25			
39	GND	–	–	Ground	GND	–
41	PER0_N	UPHY_RX8_N	D25	PCIe Receive Lane 0 (C5)	Input	PCIe
43	PER0_P	UPHY_RX8_P	D24			
45	GND	–	–	Ground	GND	–
47	PET0_N	UPHY_TX8_N	H25	PCIe Transmit Lane 0 (C5)	Output	PCIe
49	PET0_P	UPHY_TX8_P	H24			
51	GND	–	–	Ground	GND	–
53	REFCLK_N	UPHY_REFCLK3_N	F33	PCIe Reference Clock (C5)	Output	PCIe
55	REFCLK_P	UPHY_REFCLK3_P	F32			
57	GND	–	–	Ground	GND	–
59, 61, 63, 65	Key	–	–	–	–	–
67, 69	N/C	–	–	Unused	–	–
71, 73, 75	GND	–	–	Ground	GND	–
77	MNT2	–	–	Ground	GND	–
Even Pins						
2, 4	3V3	–	–	3.3V Supply	PWR	3.3V
6, 8, 10	N/C	–	–	Unused	–	–
12, 14, 16, 18	3V3	–	–	3.3V Supply	PWR	3.3V
20, 22, 24, 26, 28, 30, 32, 34, 36, 38	N/C	–	–	Unused	–	–
40	SMB_CLK	I2C3_CLK	F53	I2C 3 Clock	Bidir	1.8V (3.3V tolerant)
42	SMB_DATA	I2C3_DAT	E53	I2C 3 Data	Bidir	
44	ALERT*	GPIO58	G19	M.2 Key M Alert	Output	1.8V
46, 48	N/C	–	–	Unused	–	–
50	PERST#	PEX_C5_RST_N	L18	PCIe C5 Reset	Output	3.3V

Pin #	Connector Pin Name	Assoc. Module Pin Name	Module Pin #	Usage/Description	Type/Dir	Voltage Level
Odd Pins						
52	CLKREQ#	PEX_C5_CLKREQ_N	L19	PCIe C5 Clock Request	Bidir	3.3V
54	PEWAKE#	PEX_WAKE_N	A8	PCIe Wake	Input	3.3V
56, 58	N/C					
60,62,64,66	Key	–	–	–	–	–
68	SUSCLK	–	–	Suspend Clock	Output	3.3V
70, 72, 74	3V3	–	–	3.3V Supply	PWR	3.3V
76	MNT1	–	–	Ground	GND	–
Note: In the Type/Dir column, Output is to M.2 connector; Input is from M.2 connector. Bidir is for bidirectional signals.						

Legend	Ground	Power	Reserved/Unused
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2.6.1 Maximum M.2 Key M Card Routing Delays

The routing on the carrier board meets the requirements in the *NVIDIA Jetson Thor Series Modules Design Guide* for routing to an M.2 connector. Therefore, the maximum allowed routing on the M.2 adapter card is based on the insertion loss allocated for the card, which is 9.6 dB for Gen3, 8.0 dB for Gen4, or 9.5 dB for Gen5. Using the mid-loss PCB technology loss numbers from the design guide, this translates to the values listed in the following table.

Table 2-12. M.2 Key M Maximum Trace Delays

PCIe	Maximum Trace Delay Allowed on M.2 Card
Gen3	127 mm (average of Microstrip and Stripline guidelines)
Gen4	
Gen5	

2.7 Audio Panel Header

The carrier board includes a 10-pin (2 x 5, 2.54 mm pitch) audio panel header (J511). This can be used to connect to a standard PC audio panel to support connections to microphone, line-in, headphones, powered speakers, and so on.

Table 2-13. Audio Panel Header Pin Description

Pin #	Connector (Codec) Pin Name	Associated Module Pin Name	Module Pin #	Usage/Description	Type/Dir
1	(IN1P)	–	–	Microphone #1 input	Input
2	AGND	–	–	Ground	GND
3	(IN2P)	–	–	Microphone #2 input	Input
4	(LRCK2/GPIO4/PDM_SDA)	–	–	Presence – detects if audio dongle is connected to header.	Input
5	(HPO_R)	–	–	Headphone output right channel	Output
6	(MIC_IN_DET)	–	–	Jack/Microphone detect pin	Input
7	SENSE_SEND	–	–	Pulled to analog GND	NA
8	Key	–	–	–	–
9	(HPO_L)	–	–	Headphone output left channel	Output
10	(BCLK2/GPIO3/PDM_SCL)	–	–	Headphone or jack detection	Input
–	–	AUDIO_MCLK	H9	Audio initiator clock	Output
–	–	I2S3_SCLK	C59	I2S #3 clock	Output
–	–	I2S3_FS	C60	I2S #3 field select	Bidir
–	–	I2S3_DOUT	K59	I2S #3 data output	Output
–	–	I2S3_DIN	J59	I2S #3 data input	Input
–	–	GPIO40	C5	Audio interrupt	Output
–	–	I2C8_CLK	F52	I2C #8 clock	Bidir
–	–	I2C8_DAT	F51	I2C #8 data	Bidir

Note: In the Type/Dir column, Output is to Audio Header; Input is from Audio Header. Bidir is for bidirectional signals.

Legend	Ground	Power
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Chapter 3. Carrier Board Custom Connectors

The carrier board supports several custom expansion headers:

- > Jetson Module Connector, 65 × 11
- > CAN Header, 2 × 13, 1.27 mm pitch
- > Fan Header, 4-pin, 2 mm pitch (backup)
- > Automation Header, 2 × 6, 2.54 mm pitch
- > Button and LED Header, 2 × 3, 2 mm pitch
- > RTC Backup Battery Connector, 2-pin, 1.25 mm pitch

The routing guidelines for the interfaces supported on the expansion connectors can be found in the *NVIDIA Jetson Thor Series Modules Design Guide*. Those guidelines cover the PCB routing from the Jetson Thor to the peripheral device or actual device connector. When designing modules for one of the Jetson Thor expansion connectors, the routing on the carrier board must be accounted for. Tables are provided for the critical interfaces that provide the PCB delays on the carrier board. These delays are subtracted from the delays allowed in the *NVIDIA Jetson Thor Series Modules Design Guide* routing guidelines. The tables also include the maximum trace guidelines and remaining maximum trace delay allowed on the peripheral modules. See the *NVIDIA Jetson Thor Series Modules Design Guide* for other requirements (impedance, trace spacing, skews between signals, and so on).

3.1 Module Connector

The carrier board interfaces to Jetson Thor using a 699-pin (65 × 11) connector (J3). The part number for the connector used on the carrier board can be found in the *NVIDIA Jetson Thor Series Supported Components List (SCL)* document. This interfaces with the module. See the *NVIDIA Jetson Thor Series Modules Data Sheet* for the connector used on the module. The connector pinout can be found in the *NVIDIA Jetson Thor Series Modules Design Guide*.

3.2 CAN Header

The carrier board includes a 26-pin (2 x 13, 1.27 mm pitch) CAN header (J47). The connector used on the carrier board is a Wieson AC2800-0011-022-H0 part.

Table 3-1. CAN Header Pin Description

Pin #	Associated Module Pin Name	Module Pin #	Usage/Description	Type/Dir	Voltage Level
1	–	–	3.3V Supply (VDD_3V3)	Power	3.3V
2	–	–	5V Supply (5V_AO)	Power	5V
3	FSI_GPIO07/CAN0_DOUT	G38	CAN #0 Data Out	Bidir	Bidir
4					
5	FSI_GPIO08/CAN0_DIN	G39	CAN #0 Data In	Bidir	Bidir
6	FSI_GPIO06	J39		Input	
7	FSI_GPIO09/CAN1_DOUT	H37	CAN #1 Data Out	Bidir	Bidir
8					
9	FSI_GPIO10/CAN1_DIN	H36	CAN #1 Data In	Bidir	Bidir
10	–	–	Ground	Ground	–
11, 12, 13, 14, 15, 16, 17, 18, 19, 20	–	–	Unused	Unused	–
21	–	–	Ground	Ground	–
22, 23, 24, 25	–	–	Unused	Unused	–
26	–	–	Ground	Ground	–

Notes:

1. The module pins for the CAN ports are not directly connected to the CAN header but are routed through a transceiver.
2. In the Type/Dir column, Output is to CAN Header; Input is from CAN Header. Bidir is for bidirectional signals.

Legend

Ground	Power	Reserved/Unused
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3.3 Automation Header

The carrier board includes a 12-pin, 2 × 6 2.54 mm pitch header (J42) that makes accessible several critical system control signals. The connector used on the carrier board is a Wieson AC2100-0011-019-HH part.

Table 3-2. Automation Header Description

Pin #	Associated Module Pin Name	Module Pin #	Usage/Description	Type/Dir	Voltage Level
1	–	–	Ground	Ground	–
2	FORCE_RECOVERY_N	L10	Force Recovery Strap	Input	1.8V
3	SYS_RESET_N	L60	System Reset	Input	1.8V
4	–	–	Power Button On	Input	3.3V
5	–	–	Output from USB Type-C PD Controller (See note 1)	Output	3.3V
6	–	–	ACOK (USB Type-C or DC Jack power OK). To button power-on controller.	Input	3.3V
7	MODULE_SLEEP_N	J60	Carrier board sleep	Output	1.8V
8	SYSTEM_OC_N	A61	System Overcurrent indicator	Input	1.8V
9	UART2_CTS	A57	Wake on LAN (WOL)	Input	1.8V
10	–	–	Ground	Ground	–
11	JTAG_TRST_N	G61	JTAG Test Reset	Input	1.8V
12	–	–	Ground	Ground	–

Notes:

1. Auto-Power-On is enabled when Pin 5 and Pin 6 are tied together.
2. In the Type/Dir column, Output is to header; Input is from header. Bidir is for bidirectional signals.

Legend	Ground	Power
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3.4 Microfit3.0 Power Connector

The carrier board includes a Microfit3.0 Power Connector (J83) as an alternative to using the included USB Type-C power supply. The connector used on the carrier board is a Microfit3.0 3.0mm pitch 2 x 2 header.

Table 3-3. Microfit3.0 Power Connector Pin Description

Pin #	Usage/Description	Type/Dir
1	Option for Main DC input supplying SYS_VIN_HV	Power
2	Option for Main DC input supplying SYS_VIN_HV	Power
3	Ground	Ground
4	Ground	Ground

Legend

Ground	Power
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3.5 RTC Backup Battery Connector

The carrier board has a connector where a Real Time Clock (RTC) backup battery can be connected. This connector (J13) is commonly used on laptop designs. The connector used is a Wieson America Inc. AC2651-0011-003-HH, 2-pin, 1.25 mm pitch connector.

Table 3-4. RTC Backup Battery Connector Pin Description

Pin #	Associated Module Pin Name	Module Pin #	Usage/Description	Type/Dir
1	PMIC_BBATT	D37	Real-Time-Clock backup battery supply	Power
2	–	–	Ground	Ground

Legend

Ground	Power
--------	-------

Chapter 4. Miscellaneous

4.1 LED Indicators

Table 4-1 describe the LED indicator.

Table 4-1. LED Indicators

LED	Description	Usage
D119	SOC Regulator Power LED (Green)	Indicates when the main 5.0V (VDD_5V) supply is enabled

4.2 I2C Interface Usage

The following tables show the I2C usage on the Jetson Thor module and developer kit carrier board.

Table 4-2. Jetson Thor I2C Interface Usage

Ctrl	Module Pin Names (Thor Pins)	Usage on Module	I2C Address	On-Module Pull-up/voltage
I2C1	I2C1_CLK/DAT	ID EEPROM	7'h50	1K Ω to 1.8V
I2C2	I2C2_CLK/DAT	Current Monitors	7'h40	1K Ω to 1.8V (also current monitors via 1.8V-5V level shifters with 2.2k pull-ups to 5V)
I2C3	I2C4_CLK/DAT			1K Ω to 1.8V
I2C4	DP_AUX_CH2_P/N			
I2C6	DP_AUX_CH0_P/N			
I2C7	I2C8_CLK/DAT			1K Ω to 1.8V
I2C8	DP_AUX_CH3_P/N			
I2C9	I2C5_CLK/DAT			1K Ω to 1.8V
I2C10	DP_AUX_CH1_P/N			
I2C12	I2C3_CLK			1K Ω to 1.8V

Table 4-3. Jetson Thor Module Carrier Board I2C Interface Usage

Ctrl	Module Pin Names (Thor Pins)	Usage on Carrier Board	I2C Address	On-Board Pull-up/voltage
I2C1	I2C1_CLK/DAT	ID EEPROM	7h56	
I2C2	I2C2_CLK/DAT	USB PD Controller Power Monitor (INA238)	7'h08 7'h44	
I2C6	DP_AUX_CH0_P/N	DP Connector		
I2C7	I2C8_CLK/DAT	Audio Codec USB Hub 12V DC-DC	7'h1C 7'h6A 7'h60	
I2C9	I2C5_CLK/DAT			
I2C10	DP_AUX_CH1_P/N	HDMI Connector		10K Ω to 1.8V
I2C12	I2C3_CLK	QSFP28 M.2 Key M		

Chapter 5. Mechanicals

Figure 5-1 and Figure 5-2 show the mechanical dimensions for the carrier board.

Figure 5-1. Carrier Board Dimensions – Top View

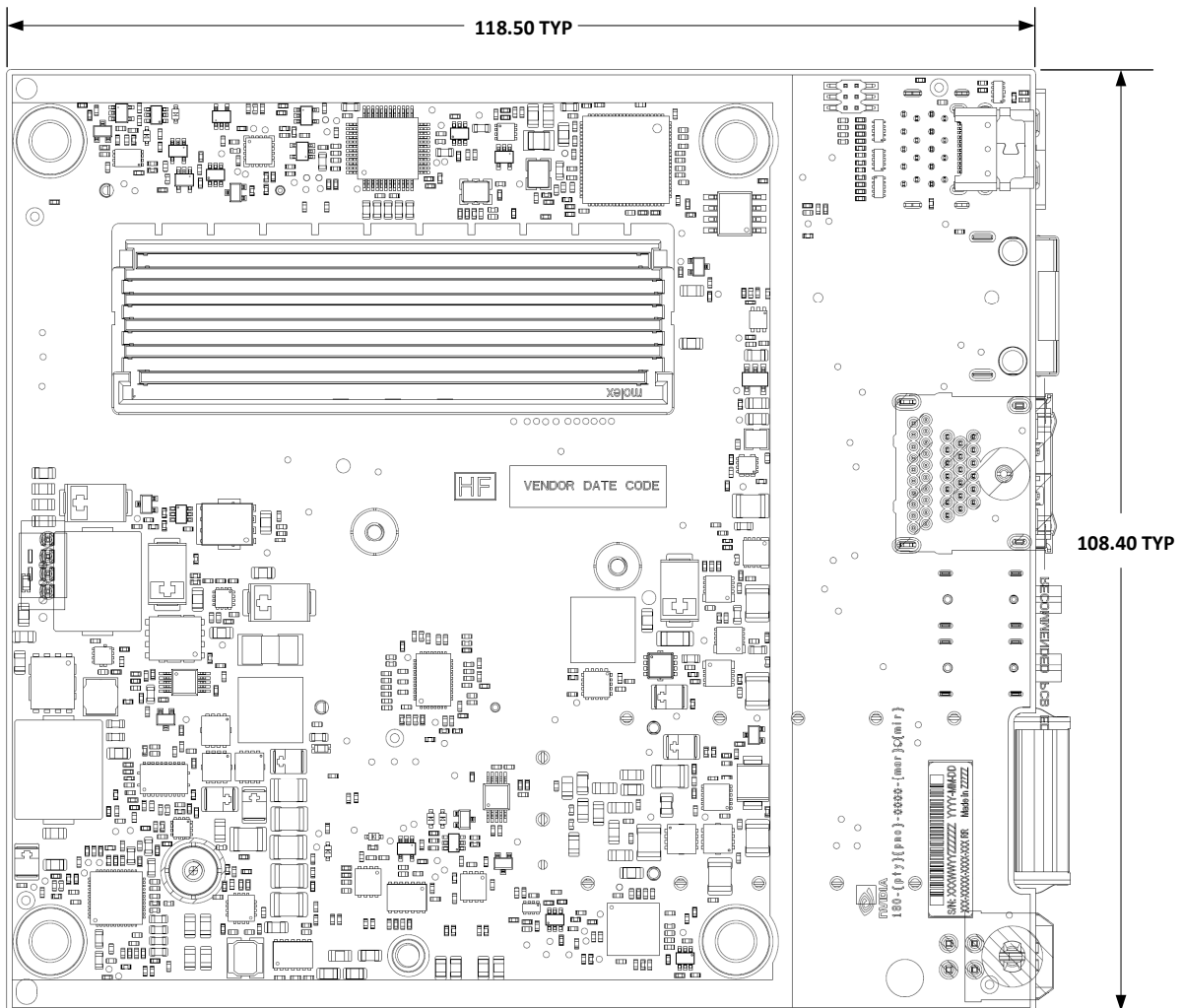


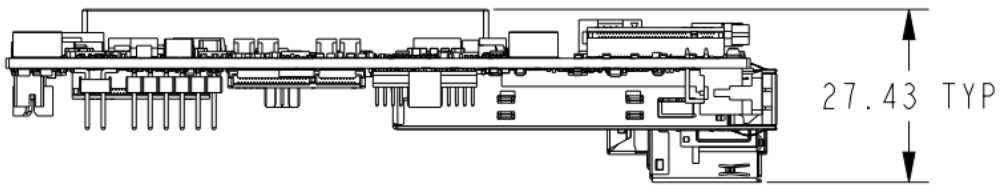
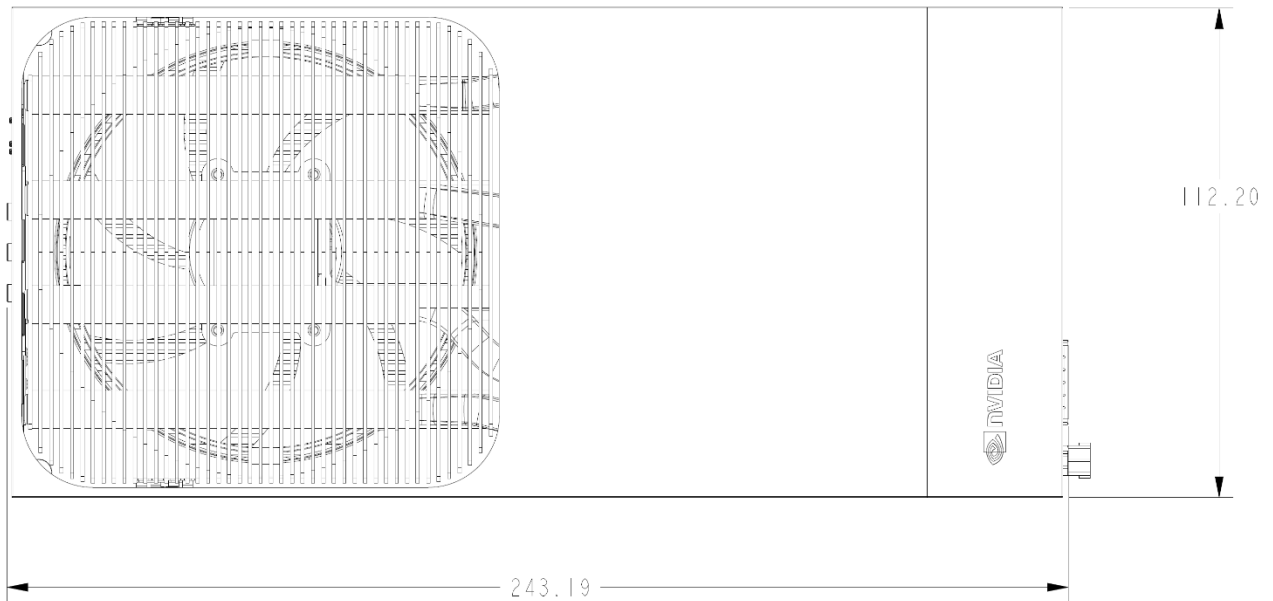
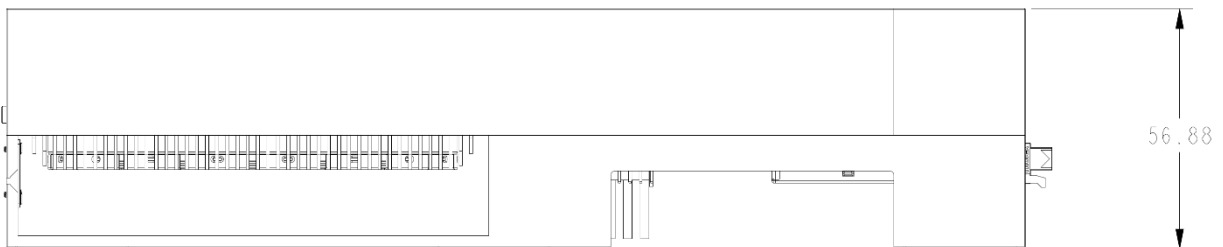
Figure 5-2. Carrier Board Mechanical Dimensions – Side View

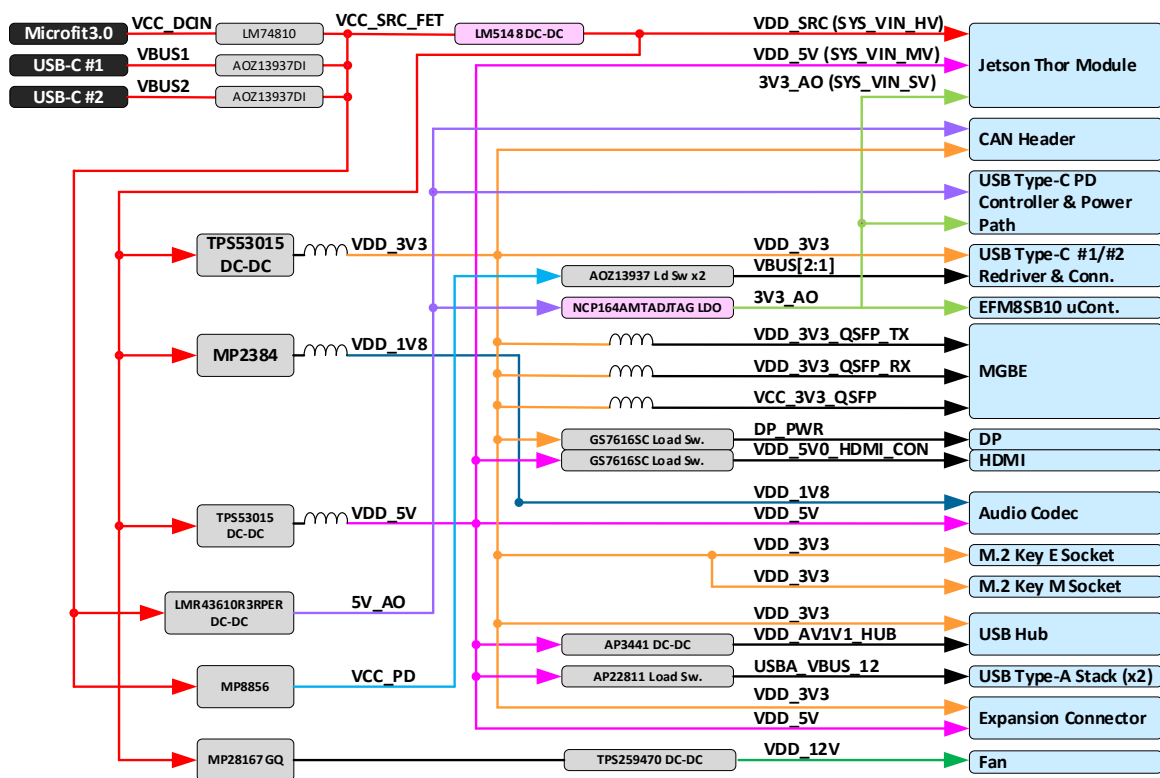
Figure 5-3 and Figure 5-4 show the mechanical dimensions for the Dev Kit. The weight of the Dev Kit is approximately 1.94kg.

Figure 5-3. Dev Kit Mechanical Dimensions – Top View**Figure 5-4. Dev Kit Mechanical Dimensions – Side View**

Chapter 6. Interface Power

The following figure shows the interface connector power diagram.

Figure 6-1. Interface Connector Power Diagram



The following tables show the allocation of supplies to the connectors on the carrier board and current capabilities.

Table 6-1. Interface Power Supply Allocation

Power Rails	Usage	(V)	Power Supply or Gate	Source	Enable
VCC_SRC	Main power input from DC Adapter or USB Type-C VBUS[2:1]	9-28	FETs	DC Adapter or USB Type-C	
VCC_PD		20	MP8856 LDO	VCC_DCIN/VBUS[2:1]	VCC_DCIN, VBUS[2:1]
5V_AO	Always-on 5V supply	5.0	LMR43610R3RPER DC-DC	VCC_SRC_FET	VCC_SRC_FET
VDD_5V	Main 5V supply	5.0	TPS53015 DC-DC	VCC_SRC	VIN_PWR_ON
VDD_3V3	Main 3.3V supply	3.3	TPS53015 DC-DC	VDD_5V	CARRIER_PWR_ON
3V3_AO	Always On 3.3V supply	3.3	NCP164AMTADJTAG DC-DC	5V_AO	5V_AO PG
VDD_1V8	Main 1.8V supply	1.8	MP2384 DC-DC	VCC_SRC	CARRIER_PWR_ON
VDD_12V	Fan	12.0	MP28167GQ-AP DC-DC	VCC_SRC	VDD_3V3
USBC[2:1]_VBUS	VBUS pin from USB Type-C connectors (alternative is DC Jack)	5-20	AOZ13937DI Ld Sw (x2)	VCC_PD	PD Controller
VDD_AV1V1_HUB	USB Hub controller rail	1.1	AP3441L	VDD_5V	VDD_3V3 PG
USBA_VBUS_12	USB Type-A Dual-port VBUS (J69)	5.0	AP22811 Load Switch	VDD_5V	Hub Power Enable
VDD_0V95_RTL	Ethernet PHY 0.95V power rail	0.95	AP3441L DC-DC	5V_AO	Ethernet PHY
VDD_1V8_RTL	Ethernet PHY 1.8V power rail	1.8	AP3441L DC-DC	5V_AO	VDD_3V3_RTL supply PG
VDD_3V3_RTL	Ethernet PHY 3.3V power rail	3.3	GS7155 LDO	5V_AO	VDD_1V8 supply PG and Automation Header
DP_PWR	3.3V rail for DP connector	3.3	GS7616SC Power Switch	VDD_3V3	Module I2S7_LRCK

Table 6-2. Interface Supply Current Capabilities

Power Rails	Usage	(V)	Max Current (A)
VCC_SRC	Main power input from USB Type-C connectors or Microfit3.0	9-28	5 (USB Type-C) 15 (Microfit3.0)
VDD_5V	Main system 5V supply	5.0	10
VDD_3V3	Main system 3.3V supply	3.3	10
VDD_1V8	Main system 1.8V supply	1.8	2
VDD_12V	12V rail for Fan	12	1.5

Notes:

1. The values shown in the “Max Current” column indicate the total power available on the expansion connectors minus the current used on the platform (not per connector pin).
2. If a given voltage rail cannot supply sufficient current for your application, a possible solution is to use a regulator to derive the required voltage from another rail that has extra headroom.

Table 6-3. Supply Current Capabilities per Connector per Supply

Power Rails	Connector	(V)	Max Allocated Current (A)
VDD_12V	Fan	12	1.5
VDD_3V3	CAN	3.3V	1
	M.2 Key M		3
	M.2 Key E		2

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